TeleScope

Customer Intelligence & Recommendation Platform

Comprehensive Academic Project Proposal & Architecture Specification

1. Project Overview & Context

The telecommunications market in Egypt features major competitive operators including Vodafone Egypt, Orange Egypt, WE, and e& Egypt. Choosing an optimal provider is challenging due to complex tariff structures and network variances. This project builds an intelligent platform combining customer feedback, surveys, and live telecom offers to generate tailored recommendations.

Core Research Question: "Which telecom provider is the best match for this customer, and why?"

2. Academic Literature Review Foundation

Grounded in academic frameworks, the project establishes context, identifies research gaps in multi-factor comparison platforms, justifies the NLP/ranking methodology, avoids redundant research, and formulates a conceptual analytical model.

3. Problem Statement & Market Scope

Customers evaluate operators across internet quality, coverage, pricing, and app usability. No single platform consolidates these into personalized rankings:

Provider	Market Category & Focus
Vodafone Egypt	Egyptian Market Leader (Data & App Experience)
Orange Egypt	Egyptian Market (Customer Service & Coverage)
WE (Telecom Egypt)	Egyptian Market (Fixed-Mobile Convergence & Bundles)
e& Egypt	Egyptian Market (Competitive Offers & Digital Growth)

4. Comprehensive Data Sources

Source Channel	Description & Data Capture Mechanism
Google Play Reviews	User reviews, ratings, and feedback harvested directly from official provider apps.
Customer Surveys	Primary quantitative questionnaires capturing user priorities, service ratings, and preferences.
Telecom Offers & APIs	Scraped and structured pricing, internet quotas, voice minutes, and validity rules.

5. End-to-End Technical Architecture (Updated Stack)

Pipeline Stage	Tools & Technologies Implemented
Data Sources	Google Play, Surveys, Websites / APIs (Python ingestion scripts)
Storage & Orchestration	Microsoft Azure Blob Storage (Raw Data Lake), Apache Airflow (DAGs)
Processing & Warehouse	PySpark (Distributed Cleaning), PostgreSQL
ML & Intelligence Layer	Sentiment Analysis, Hybrid Recommender (rule-based filters, then ML ranking)
Presentation Layer	Streamlit (Interactive Frontend UI) with a Python services layer, Power BI

Recommendation method. Hybrid recommender: Step 1 applies rule-based filters for hard constraints (budget, quota, validity); Step 2 applies ML ranking to the eligible offers using validated features.

6. Expected Project Outcomes

The final platform delivers actionable market intelligence, automated provider comparisons, sentiment diagnostics, and interactive user recommendations via Streamlit.